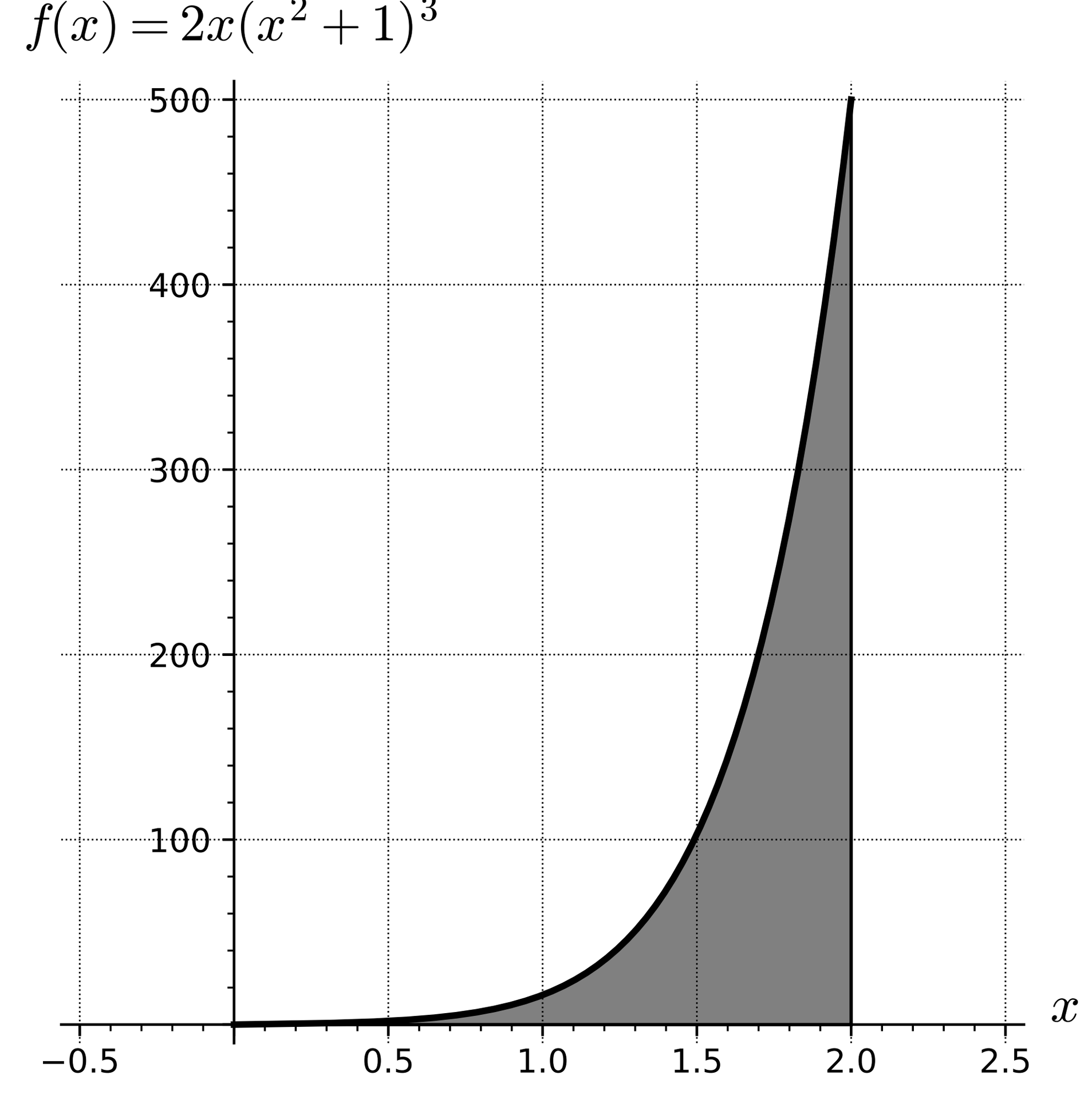$f(x) = 2x(x^2+1)^3$